

2022

Time : 3 hours

Full Marks : 70

*Candidates are required to give their answers
in their own words as far as practicable.*

The figures in the margin indicate full marks.

Answer all sections as directed.

Section-A

(Compulsory)

1. Pick up the correct alternative for each of
the following questions : $2 \times 10 = 20$

(a) If $A = B = C = 1$, then the value of
 $A \oplus B \oplus C$ is equal to :

- (i) 1
 - (ii) 2
 - (iii) 0
 - (iv) None of these
- (b) If $S = R = 1$, in SR flip flop then the next state is :
- (i) Predictable
 - (ii) Unpredictable
 - (iii) Complement of previous state
 - (iv) None of these
- (c) The value of $(A \oplus B)'$ is equal to :
- (i) $AB + A'B'$
 - (ii) $A \odot B$
 - (iii) Both (i) & (ii)
 - (iv) None of these

(d) Sum of all literals is known as :

- (i) Minterms
- (ii) Maxterms
- (iii) Both (i) & (ii)
- (iv) None of these

(e) D-flip flop is obtained from :

- (i) SR flip flop
- (ii) JK flip flop
- (iii) Master slave flip flop
- (iv) None of these

(f) $F = \prod(1,3,4,5)$ is equivalent to :

(i) $F = M_1 + M_3 + M_4 + M_5$

~~(ii)~~ $F = m_1 + m_3 + m_4 + m_5$

(iii) $F = M_1 \cdot M_3 \cdot M_4 \cdot M_5$

(iv) $F = m_1 \cdot m_3 \cdot m_4 \cdot m_5$

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(g) An encoder 2^n has input lines then the output lines is :

- (i) $n+1$
- (ii) n
- (iii) $n-1$
- (iv) None of these

(h) Prefix notation means :

- (i) Places the operator before operands
- (ii) Place the operator after operands
- (iii) Places the operator between operands
- (iv) None of these

(i) Specify the operand in the instruction is called :

- (i) Immediate addressing mode
 - (ii) Register addressing mode
 - (iii) Direct addressing mode
 - (iv) None of these
- (j) Subtraction in computers is carried out by :
- (i) 1s complement
 - (ii) 2s complement
 - (iii) 9s complement
 - (iv) 3s complement

Section-B

(Short answer type questions)

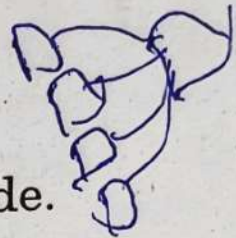
Answer any **four** questions of the following :

$$5 \times 4 = 20$$

2. Differentiate between combinational and sequential circuits.

3. Prove the De Morgan's theorem
 $(A + B)' = A'.B'$ by algebraic method.
4. Draw the circuit diagram for the following boolean expressions :
- (a) $F = (AC)' + (AB)' + (BC)'$
- (b) $F = (A + C)' . (A + B)' . (B + C)'$
5. Define the following terminologies :
- (a) Logic gates
- (b) Decoder
- (c) Multiplexer
- (d) Post fix notation
- (e) Full Adder
6. Convert the following numerical arithmetic expression into postfix notation and show the

stack operation for evaluating the numerical result $(7-5) * (8/2)$.



7. Write short notes on direct addressing mode.

Theory The

Section-C

(Long answer type questions)

Answer any two questions of the following :

$$15 \times 2 = 30$$

8. Represent 500.3 using the IEEE 754 standard for 32-bit floating point representation.
9. Convert the following into canonical POS form : $A'B + B(C'(C' + B))$
10. Explain SR flip flop with the help of graphic symbol and characteristic table. Write the disadvantages of SR flip flop. Derive excitation table from SR characteristics table.

11. Simplify the following boolean function F together with the don't care condition d in

(i) Sum of products form

(ii) Product of sum form

$$F(W,X,Y,Z) = \sum(1,3,7,11,15)$$

$$d(w,x,y,z) = \sum(0,2,5)$$

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